# 5776K'S PNEUMATICS

applications to VEX Over Under (23-24)

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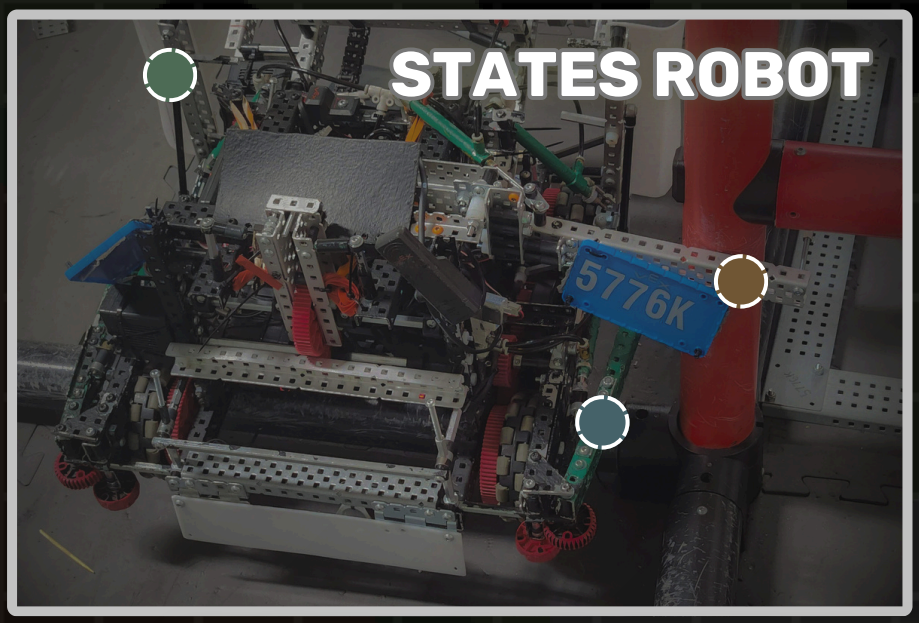
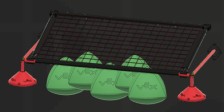
*KIWI KREW* CAPTAIN

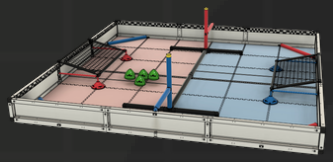



## VEX OVER UNDER CONTEXT

*1. **triballs** scored in the **goal** are worth 5 points*



*2. **triballs** in the **offensive zone** are worth 2 points*



*3. **elevation** - there are 11 tiers that a robot may be hung at ranging from A-J (A is the lowest)*

- *points are **RELATIVE** to other robot tiers*

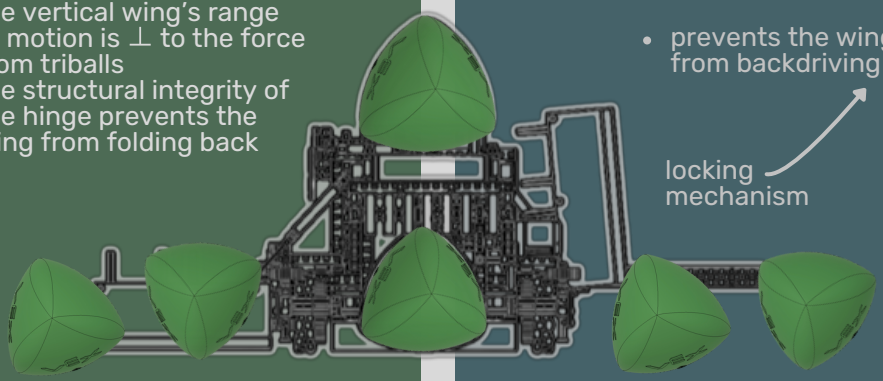
## PNEUMATICS SUBSYSTEM BREAKDOWN

### VERTICAL WINGS

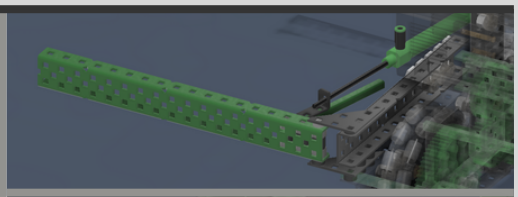
### HORIZONTAL WINGS

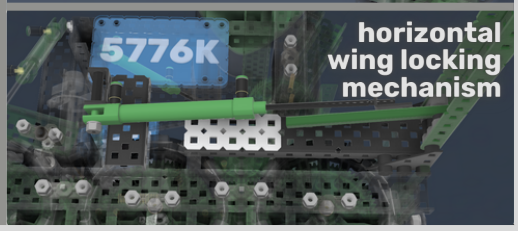
*WINGS: channels that extend out up to the 36" width limit using a piston actuation to push tribals on the field either 1) scoring into the goal or 2) across the barrier into the home offensive zone*

- the vertical wing's range of motion is ⊥ to the force from triballs
- the structural integrity of the hinge prevents the wing from folding back
- prevents the wing from backdriving




STATES DESIGN

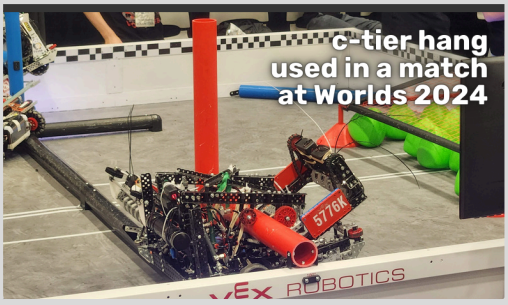





### ELEVATION

***much of the season had robots elevating at A tier using a "balance" mechanism*

STATES DESIGN CAD

we found 5 extra points not worth developing a hanging mechanism when we could allocate more time to driving practice and programming refinement

CURRENT PISTON GEOMETRY